

A reduced cluster dynamics modeling of radiation damage in tungsten

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ABSTRACT

A reduced cluster dynamics model is developed to investigate the microstructure evolution of irradiated single crystal tungsten. The model accounts for the generation and reaction of point defects, small-size defect clusters, helium clusters, as well as the nucleation and growth of large immobile defects, including the interstitial dislocation loops, voids and helium bubbles. Moreover, by incorporating an atomically informed loop punching mechanism for bubble growth, the model is able to accurately capture the evolution kinetics of radiation-induced defects with and without the helium implantation. The predicted densities and sizes of loops and voids/bubbles, the helium-to-vacancy ratio and the internal pressure of helium bubbles are all in good agreement with experimental data at different irradiation doses across a wide range of temperatures from 300 K up to 1200 K. This work aims to provide a robust tool for analyzing the microstructure of irradiated tungsten under both fission and fusion conditions.

1. Introduction

In the context of growing global energy demands and increasing environmental concerns, nuclear fusion energy stands out as a promising future energy solution due to its cleanliness, efficiency, and sustainability. Due to the very high working temperature in fusion reactors, tungsten-based materials, which have ultra-high melting point, excellent thermal properties, low hydrogen solubility, and low sputtering yield [1,2], are considered as the candidate materials for the plasma facing components (PFC), including the first wall and the divertor. In fusion reactors like Tokamak, these plasma-facing materials withstand the impacts of high-energy neutrons and hydrogen isotopes (H, D, and T) as well as helium (He) ions produced by the D-T fusion reaction [3], leading to severe damage on the material surface and its interior. In particular, helium tends to combine with vacancies to form helium bubbles [4], which in turn enhance vacancy retention, result in surface fuzzing [5], and cause the detrimental effects on materials' mechanical properties such as hardening and embrittlement. Furthermore, the density of helium bubbles formed due to helium implantation is significantly higher than voids (without helium), leading to severer

degradation. It is therefore important to investigate the formation, and the evolution of radiation induced defects for plasma-facing materials.

Over the past few decades, extensive experiments have been conducted to study the microstructural evolution of tungsten under neutron irradiation. For instance, Hasegawa et al. [6] conducted experiments with different neutron irradiation sources and observed various types of defects at many temperatures and dpa levels. Hu et al. [7] summarized recent experiments in terms of the size and density of voids and interstitial loops in tungsten and tungsten alloys at different radiation conditions. Fukuda et al. [8] studied the density and size of voids and loops of irradiated W-Re alloys with varying Re content. Their experiments discovered that voids do not form at low temperatures irrespective of Re concentrations. However, these experiments were tested without helium implantation. On the other hand, ion irradiation experiments with helium are commonly used to emulate fusion irradiation conditions. By using 6.4 MeV Fe³⁺ and 1 MeV He⁺ dual-beam irradiation, Zhang et al. [9] found that the helium bubbles were formed only above 500 °C, while dislocation loops were observed at both low and high temperatures. With 10 keV He⁺ irradiation sources, Yi et al. [10] revealed that the growth kinetic of helium bubble was strongly affected by temperatures,

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whereas the He to vacancy ratio stabilized between 2 and 3 in the temperature range of 1073–1473 K [10]. Similarly, Ipatova et al. [11] observed that the size and density of helium bubbles increased with irradiation dose at 800 °C. In addition, Ding et al. [12] performed H_2^+ and He^+ dual-beam irradiation experiments, obtaining the relationship between the size and density of dislocation loops and bubbles with irradiation dose.

To better understand microstructural evolution under irradiation conditions, various multiscale simulation methods have been adopted over the decades as well. For instance, molecular dynamics (MD) is widely used to simulate irradiation cascades [13], and to study defect energies [14] and the mechanism of defect nucleation and growth [15,16]. Object Kinetics of Monte Carlo (OKMC) can cope with the migration and interaction of individual radiation-induced defects, which has been successfully applied to investigating the post-cascade defect recombination [17] and the helium bubble coalescence [5]. However, these simulations are always limited by their length and time scales, and the investigated radiation dose (<0.1 dpa) is far below the fusion condition. Under this circumstance, cluster dynamics (CD) is an appropriate alternative, which is on the basis of the mean-field approximation and is able to modelling the microstructure evolution up to 100 dpa [18].

Cluster dynamics primarily relies on the mean field theory approach, where all spatial correlations between the clusters of defects are ignored. This method has been successfully used to study the evolution of point defects [19], void swelling [20,21], and irradiation creep [22] in various materials. As an example of tungsten, Ghoniem et al. [23] developed a cluster dynamics model accounting for the helium defects, and studied the defect evolution in tungsten with helium implantation. It was found that the helium migration rate is related to the size of helium-vacancy clusters with different mechanisms. And stemming from a variety of situations where interfaces play a significant role in the evolution of the defect microstructure, spatially resolved CD was developed [24] to predict the spatial distribution of defects. Recently, Li et al. [25] developed a spatially resolved cluster dynamics model to investigate the accumulation and diffusion of helium atoms in W under helium/neutron co-irradiation conditions. By using this model, the concentration profiles of different defects along with the depth have been examined for different He ions incident conditions. With the development of MD simulations, it was further observed that cascades generate not only point defects but also small clusters [26], which can undergo one-dimensional or three-dimensional migration [27]. This led to the integration of the Production Bias Model (PBM) [28] into cluster dynamics, which accounts for defect cluster generation biases and different cluster migration modes. With considering the production bias, Shah et al. [18] successfully predicted the elastic stored energy in irradiated tungsten film using a spatially resolved CD model. In addition, cluster dynamics is also coupled with other microscopic simulation methods to provide a more comprehensive investigation of the defect reaction process. Using coupled CD-OKMC, Edward et al. [5] studied the nucleation and migration of helium bubbles and highlighted the effect of the vacancy trap mutation on the bubble growth.

However, it should be noted that most of these CD models require heavy computations for solving a large number of ordinary (or partial) differential equations, i.e. one for each defect. Moreover, they assumed that all mobile defects present 3D diffusion, ignoring the 1D migration feature of small SIA clusters, which may result in a significant deviation of the defect evolution from the real case [29]. To address these limitations, we develop a reduced cluster dynamics model, which simplifies the usual hierarchy of cluster dynamics equations by focusing on the stability of small defect clusters once nucleated. This type of model considers both 1D and 3D migrating defect and has been successfully applied to studying many long-term irradiation phenomenon, e.g., the irradiation growth in zirconium [30,31] and the defect evolution in iron [32]. In addition, some experiments [33] and molecular dynamics simulations [16] have shown that the loop punching mechanism plays a crucial role in the bubble nucleation and growth, which may be the key

growth mechanism for bubbles with high internal pressure. Accordingly, we introduce such atomic simulation informed loop punching mechanism into our CD model and thoroughly discusses its impact on bubble kinetics in irradiated tungsten.

The paper is organized as follows. A general model description is presented in the next section, including the key assumptions, governing equations and model parameters. The microstructure evolution of irradiated single crystal tungsten without and with helium implantation are exhibited and compared to the experimental data in section 3.1 and 3.2, respectively. The effects of the loop punching mechanism are further discussed in subsections 3.3 and 3.4. Finally, conclusions of the present work are shown in section 4.

2. Model description

2.1. Defect evolution process and basic assumption

Under irradiation, the incident high-energy particles induce collision cascades within the materials, leading to the generation of Frenkel pairs, vacancy and interstitial clusters. Point defects generally perform three-dimensional isotropic diffusion, while small clusters migrate one-dimensionally with relatively higher mobility. As irradiation progresses, these mobile defects diffuse and interact with each other, resulting in the agglomeration of large-size defects, e.g., dislocation loops, stacking fault tetrahedrons and voids, depending on materials property, radiation dose and temperature [34]. As for tungsten, a typical body-centered cubic metal, the commonly observed radiation-induced defects are dislocation loops and voids. In conjunction with the helium implantation, the injected helium tends to combine with vacancies, which leads to the formation of helium bubbles [35,36]. After nucleation, helium bubbles grow by absorbing vacancies and helium atoms, while interstitial-type clusters absorb vacancy or interstitial defects only. Accordingly, the present work primarily considers the migration and interaction of point defects, small mobile defect clusters, and the nucleation and growth of immobile interstitial loops, and voids/bubbles for irradiated single crystal tungsten. In the developed cluster dynamics framework, all types of radiation-induced defects are divided into two groups, i.e., small mobile defect clusters and large immobile defect sinks. For small defect clusters, their concentration is described by individual rate equations, while for defect clusters with a size larger than a critical value, we treat them as immobile sinks, characterized by their number density and average size. This method can significantly reduce the number of equations of the system, and at the same time accurately capture the features of the microstructure evolution of irradiated materials. One vital challenge of this model is to determine the critical nucleation size, above which we do not trace their individual evolutions.

Because the binding energy characterizes the stability of the clusters, the criterion for such simplification is based on this energy barrier (E_b). As shown in Fig. 1(b), it is obvious that the critical helium to vacancy ratio of He-V clusters is 2, where the binding energy of helium atoms with a helium-vacancy cluster equals to that of vacancies. We have checked that including the higher hierarchy of helium-vacancy complex presents little influence on the results. We therefore make the simplest assumption that the helium vacancy cluster He_2V is the critical nucleation size as in the previous models [23,37]. In the present model, the mixed helium-interstitial clusters He_nI_k are ignored since their binding energy is far below the value of their helium-vacancy cluster counterparts $E_{He_nV_m}^b$ [36]. Then, as the migration energy reduces with the increase of the cluster's size, we further assume that SIA clusters with sizes larger than 3 ($I_k, k > 3$) are immobile SIA loops, and vacancy clusters ($V_m, m > 1$), helium clusters ($He_n, n > 2$) and helium-vacancy clusters ($He_nV_m, n > 2, \text{ or } m > 1$) with sizes larger than their critical values are voids or bubbles as summarized in Fig. 1(a). Moreover, the mobile defects in the model consists of 3D diffusional SIA(I), single vacancies(V), single helium atoms (g), di-helium (2 g), and 1D diffusional di-interstitials (I_2)

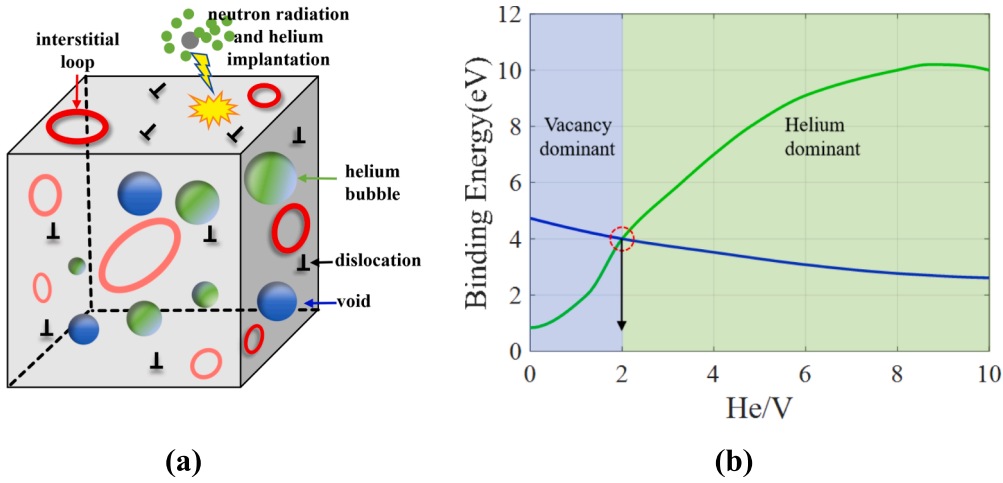


Fig. 1. (a) Major radiation-induced defects of the model. (b) Binding energy of a helium atom (green line) and a vacancy (blue line) with a mixed He-V cluster as a function of the helium to vacancy ratio. (Reproduced from Shah [18]).

and tri-interstitials (I_3).

Actually, large SIA clusters as well as prismatic dislocation loops are all mobile, which can migrate one-dimensionally along their Burgers vector [38,39]. However, the glide motion of dislocations is much faster than the diffusion-controlled climb motion, hence a self-organized dislocation loop pattern is rapidly formed by dislocation glide. Once the loops aggregate and raft, their elastic reactions will prevent the further motion of loops. It is therefore rational to assume that larger SIA clusters are immobile dislocation loops. Moreover, mean field theory based reduced cluster dynamics ignores all spatial correlations of defects, and the defects are simply characterized by their average sizes and densities. At relatively low dose where the average loop size is much smaller than the loop spacing, the loop coarsening is also neglected. On the other hand, recent MD [40] and DFT [36] calculations have shown that helium clusters with sizes smaller than 6 are capable to migrate. Based on the diffusion coefficients of helium clusters reported in Shah et al. [18] however, it is apparent that He and He₂ have the largest diffusivity and thus are dominant species during reactions. As for helium-vacancy clusters, Perez et al. [41] suggests that the mobility of helium-vacancy clusters is primarily dependent upon the number of helium atoms and their diffusion coefficients are several orders of magnitude smaller than those of helium clusters. We therefore assume that larger helium clusters and helium-vacancy clusters are all immobile.

In summary, the migration and interaction of these mobile defects result in the nucleation of immobile defects, including interstitial loops and voids/bubbles. The later acts as sinks to 1-D and 3-D migrating defects, leading to the growth or shrinkage of immobile defects, and thus the macroscopic change of the tungsten's properties.

2.2. Cluster dynamics equations

The cluster dynamics model is based on the mean-field rate theory, which assumes that all defects and sinks are homogeneously distributed in space and that each position receive an identical defect flux [21,42,43]. By considering the diffusion of mobile defects and the possible reaction with other defects, the concentration evolution of point defects (vacancies and interstitials) and small defect-clusters (di-interstitials, tri-interstitials, helium atoms and He-V clusters) is described by following equations:

$$\frac{dC_i}{dt} = \frac{G}{\Omega} (1 - \varepsilon_{2i} - \varepsilon_{3i}) + \beta_{2i}^v C_v C_{2i} + 2\alpha_{2i} C_{2i} + \alpha_{3i} C_{3i} - R_{iv} C_i C_v - \beta_i^i C_i C_i - \beta_{2i}^i C_i C_{2i} - \beta_{3i}^i C_i C_{3i} - \beta_{gv}^i C_i C_{gv} - \beta_{2gv}^i C_i C_{2gv} - \rho_{sink}^i D_i C_i \quad (1)$$

$$\frac{dC_{2i}}{dt} = \frac{G}{\Omega} \frac{\varepsilon_{2i}}{2} + \beta_{3i}^v C_v C_{3i} + \alpha_{3i} C_{3i} + \frac{1}{2} \beta_i^i C_i C_i - \alpha_{2i} C_{2i} - \beta_{2i}^i C_i C_{2i} - \beta_{2i}^v C_v C_{2i} - k^2 D_{2i} C_{2i} \quad (2)$$

$$\frac{dC_{3i}}{dt} = \frac{G}{\Omega} \frac{\varepsilon_{3i}}{3} + \beta_{2i}^i C_i C_{2i} - \alpha_{3i} C_{3i} - \beta_{3i}^v C_v C_{3i} - \beta_{3i}^i C_i C_{3i} - k^2 D_{3i} C_{3i} \quad (3)$$

$$\frac{dC_v}{dt} = \frac{G}{\Omega} (1 - \varepsilon_{2i}^{eff}) + \alpha_{gv} C_{gv} + \alpha_{2gv}^v C_{2gv} + \delta_{gv} C_{gv} + \delta_{2gv} C_{2gv} - R_{iv} C_i C_v - \beta_{2i}^v C_v C_{2i} - \beta_{3i}^v C_v C_{3i} - \beta_{gv}^v C_v C_{gv} - \beta_{2gv}^v C_v C_{2gv} - \beta_{gv}^v C_v C_{gv} - \beta_{2gv}^v C_v C_{2gv} - \rho_{sink}^v D_v C_v + 4\pi r_b N_b D_v C_v^e \left\{ \exp \left[\frac{\Omega}{k_B T} \left(\frac{2\gamma_b}{r_b} - p \right) \right] \right\} \quad (4)$$

$$\frac{dC_g}{dt} = \frac{G_g}{\Omega} + 2\alpha_{2g} C_{2g} + \alpha_{gv} C_{gv} + \alpha_{2gv}^g C_{2gv} + \delta_{gv} C_{gv} + 2\delta_{2gv} C_{2gv} + 2\delta_{2g} C_{2g} + \beta_{gv}^g C_i C_{gv} - \beta_{gv}^g C_g C_{gv} - \beta_{2gv}^g C_g C_{2gv} - \beta_{gv}^g C_v C_g - \beta_{2g}^g C_g C_{2g} - \beta_{2g}^g C_g C_{2g} - \rho_{sink}^g D_g C_g \quad (5)$$

$$\frac{dC_{2g}}{dt} = \alpha_{2gv}^g C_{2gv} + \frac{1}{2} \beta_{gv}^g C_g C_g + \beta_{2gv}^i C_i C_{2gv} - \alpha_{2g} C_{2g} - \delta_{2g} C_{2g} - \beta_{2g}^g C_g C_{2g} - \beta_{2g}^v C_v C_{2g} - \beta_{2g}^{2g} C_{2g} C_{2g} - \beta_{2gv}^{2g} C_{2g} C_{2gv} - \rho_{sink}^{2g} D_{2g} C_{2g} \quad (6)$$

$$\frac{dC_{gv}}{dt} = \alpha_{2gv}^g C_{2gv} + \beta_{gv}^g C_v C_g - \alpha_{gv} C_{gv} - \delta_{gv} C_{gv} - \beta_{gv}^g C_g C_{gv} - \beta_{gv}^v C_v C_{gv} - \beta_{gv}^i C_i C_{gv} - \beta_{2gv}^g C_{2g} C_{gv} \quad (7)$$

$$\frac{dC_{2gv}}{dt} = \beta_{2g}^v C_v C_{2g} + \beta_{gv}^g C_g C_{gv} - (\alpha_{2gv}^v + \alpha_{2gv}^g) C_{2gv} - \delta_{2gv} C_{2gv} - \beta_{2gv}^v C_v C_{2gv} - \beta_{2gv}^i C_i C_{2gv} - \beta_{2gv}^g C_g C_{2gv} - \beta_{2gv}^{2g} C_{2g} C_{2gv} \quad (8)$$

where $G = G^{NRT} f$ and G^{NRT} is the standard dose rate, f being the surviving fraction, which characterizes net defects generated during the radiation cascade. Ω is the atomic volume and is assumed to be the same for SIAs, vacancies or helium atoms for simplicity. ε_{2i} , ε_{3i} are the fractions of interstitial clusters produced in the form of di- and tri-

interstitials, respectively. $\varepsilon_v^{\text{eff}} = \varepsilon_v(1 - V_{\text{cap}}N_b)$ is the effective fraction of vacancy clusters directly generated from the cascade, which depends on the void/bubble number density and the saturation volume [30]. $G_g = \Omega(A \times \phi_{\text{He}})$ is the helium implantation rate, where A represents the production probability and ϕ_{He} is the ion flux. Moreover, the last term in Eq. (4) refers to vacancies that are lost due to the evaporation of helium bubbles from their surfaces, which is characterized by the vacancy equilibrium concentration $C_v^e = \exp(-E_v^f/k_B T)$. Other reaction coefficients are discussed in more detail as follows:

First, the recombination rate of single SIAs and single vacancies is given as $R_{iv} = 4\pi r_{iv}(D_i + D_v)$ [43–45], where r_{iv} represents the recombination radius. And D_k is the diffusion coefficient of k -type monomers with $D_k = D_{0,k} \exp(-E_k^m/k_B T)$ [46–48], where $D_{0,k}$ is the diffusion pre-exponential factor and E_k^m is its migration energy, k_B being the Boltzmann constant and T is temperature in Kelvin.

Second, α and β in above equations represent the emission and reaction rates involving mobile defects, which depends on the stability and

mobility of defect clusters in the material. Specifically, $\alpha_k^j = \beta_{k-j}^j \exp\left(-\frac{E_k^b}{k_B T}\right)/\Omega$ is the j -type defect ($j = i, v, g$) emission frequency of the k -type

cluster ($k = 2i, 3i, 2g, gv, 2gv$), where E_k^b the corresponding bind energy of the k -type cluster. On the other hand, β depends on the shape and migration pathway of two interacting clusters [32]. The expressions of reaction rate and sink strength are summarized in Fig. 2(a). Actually, the reaction rates can be decamped into two parts as shown in [30,49], i.e., $\beta_P^Q = \beta_P^{Q(im)} + \beta_P^{Q(im)}$, corresponding to the reaction rate between the immobile defect P ($P = i, 2i, 3i, v, g, 2g, gv, 2gv$) and the mobile defect Q ($Q = i, v, g, 2g$), as well as the reaction between the immobile defect Q and the mobile defect P . Accordingly, for the case of 3D migrating clusters P interacting with 3D migrating clusters Q , the total reaction rate yields $\beta_P^Q = 4\pi(r_P + r_Q)(D_P + D_Q)$ [30,45–48,50,51]; And for the case of 1D migrating SIA clusters P interacting with 3D migrating clusters Q , the total rate is $\beta_P^Q = 2\pi(r_P + r_Q)D_Q + (\pi(r_P + r_Q)^2)^2 D_P C_Q$ [24,30,32,52]. The point defects and small clusters are described as

spheres with radii $r_i = \left(\frac{3\Omega}{4\pi}\right)^{\frac{1}{3}}$, $r_{ngmv} = \left(\frac{3\Omega(n+m)}{4\pi}\right)^{\frac{1}{3}}$, here $n = 0, 1, 2$ and $m = 0, 1$, while the radii of small planar clusters are written as $r_{ni} = \left(\frac{n\Omega}{\pi b}\right)^{\frac{1}{2}}$, here $n = 2, 3$. It is worth noting that due to low reaction probability, we ignore reactions between 1-dimensional migrating clusters.

Third, it is observed that helium clusters can dissociate under the

synergistic effects of neutrons and helium irradiation, which is known as the resolution mechanism [18,53–55]. In Eqs. (4)–(6) $\delta_{ngmv} = \sigma_s(a, \text{He})\phi(E_n)$, ($n = 1, \text{or}, 2, m = 0, \text{or}, 1$) thus characterizes the probability per second of such resolution events, where, $\sigma_s(a, \text{He})$ is the neutron scattering cross-section for helium, and $\phi(E_n)$ is the flux of neutron with an energy of E_n .

Next, sink strengths shown in Fig. 2(b) are another most important parameters of the model, which characterize the loss rate of mobile defects to different types of microstructural sinks. For instance, the sink strength ρ_{sink}^j represents the inverse of the square of the mean free path traveled by the mobile defects before it is absorbed by a stationary sink [24]. According to the isotropic diffusion model and the superposition principle, the total sink strength to mobile defects that migrate in 3D is [24,32,52]

$$\rho_{\text{sink}}^j = \rho_d Z_d^j + \rho_{il} Z_{il}^j + \rho_b Z_b^j = \rho_d Z_d^j + 2\pi r_{il} N_{il} Z_{il}^j + 4\pi r_b N_b Z_b^j, \quad (9)$$

$j = i, v, g, 2g$

where ρ_d, ρ_{il} and ρ_b represent the individual sink strength of dislocation lines, interstitial loops and bubbles/voids to the mobile defect of type j . r_{il}, r_b, N_{il} and N_b are the radius of interstitial loops, the radius of bubbles, the number density of interstitial loops and the number density of bubbles. Z_i^j corresponds to the bias factor of i -type sink to defect j . As for 1D migrating defects, the total sink strength is given as [52]

$$k^2 = 2 \left(\frac{\pi r_{cd}}{2} \left(\frac{\rho_d}{m} + \frac{2\pi r_{il} N_{il}}{m} \right) + \pi r_b^2 N_b \right)^2 \quad (10)$$

where r_{cd} is the capture radius, and m is a crystal structural dependent factor. It is seen that the total sink strength consists of two terms, i.e., the partial sink strength of the dislocation lines/loops and the partial sink strength of bubbles/voids. And because of the one-dimensional motion of the clusters, only a portion of dislocation lines/loops whose Burgers vector is parallel to the migration direction can absorb these defects. There are 4 close-packed directions in the bcc metal, so m is set to be 4.

In the following, we present the equations describing the nucleation and growth of dislocation loops and bubbles. Assuming that $I_k (k > 3)$, $V_m (m > 1)$, $\text{He}_n (n > 2)$ and $\text{He}_n V_m (n > 2, \text{or}, m > 1)$ are large immobile sinks, and grouping vacancy clusters, helium clusters and helium-vacancy clusters as a single type of sink, their nucleation rates are simply written as:

$$\frac{dN_{il}}{dt} = \beta_{3i}^i C_i C_{3i} + \Gamma_p N_b \quad (11)$$

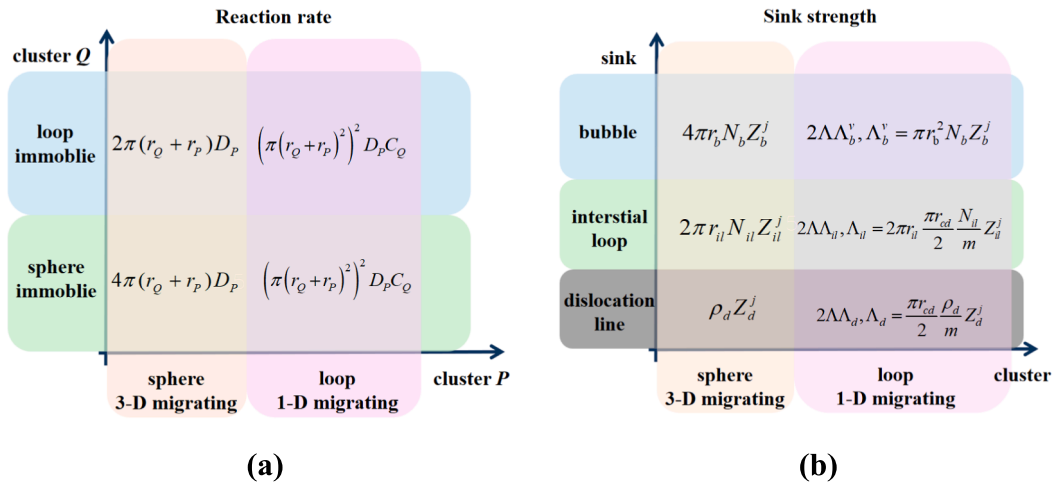


Fig. 2. (a) Summary of reaction rates between a mobile cluster P and an immobile cluster Q . (b) Partial sink strengths of immobile defect cluster to various mobile defects. The total sink strength is simply the sum of partial strengths of different sinks in terms of dislocation lines, interstitial loops and bubbles.

$$\frac{dN_b}{dt} = \frac{G_{Ev}}{4\pi r_{nuc}^3} (1 - V_{cap} N_b) + \beta_{gv}^v C_v C_{gv} + \beta_{2gv}^v C_v C_{2gv} + \beta_{2g}^g C_g C_{2g} + \beta_{gv}^{2g} C_{2g} C_{gv} + \beta_{2gv}^g C_g C_{2gv} + \beta_{2g}^{2g} C_{2g} C_{2gv} + \frac{1}{2} \beta_{2g}^{2g} C_{2g} C_{2g} \quad (12)$$

It is seen that Eq. (11) includes two terms as SIA loops can be formed by both diffusion reaction of mobile SIAs and the loop punching, where Γ_p is the average emission rate of a single bubble, it will be discussed in details in section 3.3. In regards to vacancy type sinks, they are formed either by diffusion reaction of vacancies and helium atoms, or directly from cascade collisions with the critical nucleation size r_{nuc} and the reciprocal of saturation density $V_{cap} = 1/N_{sat}$ [30]. As a large number of experiments show that the defect density increases linearly with the decrease of temperature [56–58], the saturation density of voids/bubbles can be therefore written as [59]:

$$N_{sat} = N_{sat0} (P_{T1} + TP_{T2}) \ln \left(e + P_{G_g} \frac{G_g}{G_{NRT}} \right) \quad (13)$$

where N_{sat0} is the reference saturation density, and P_{T1}, P_{T2} characterize the temperature dependence of the saturation density. Similarly, P_{G_g} is a helium-implantation-dependent parameter and $\frac{G_g}{G_{NRT}}$ represents the ratio of helium production rate to vacancy production rate, which affects the density of helium bubbles as shown in experiments [60].

Interstitial loops and bubbles are regarded as sinks that absorb mobile defects, leading to their growth or shrinkage. Hence, their size can be characterized by the total number of defect atoms (per lattice site) in the corresponding sinks, i.e. S_i for interstitial loops and S_b for voids/bubbles respectively. Their evolution can be simply computed from the net flux of the mobile defects towards the sinks as [30,32,61,62]:

$$\frac{dS_i}{dt} = \Omega [2\pi r_{il} N_{il} (D_i C_i Z_{il}^i - D_v C_v Z_{il}^v) + 2k_{il}^2 D_{2i} C_{2i} + 3k_{il}^2 D_{3i} C_{3i}] + \pi \left(\frac{\sqrt{2}}{2} r_b \right)^2 b \Gamma_p N_b \quad (14)$$

$$\frac{dS_b}{dt} = \pi \left(\frac{\sqrt{2}}{2} r_b \right)^2 b \Gamma_p N_b - 2\Omega k_b^2 D_{2i} C_{2i} - 3\Omega k_b^2 D_{3i} C_{3i} + 4\Omega \pi r_b N_b \left(D_v C_v + D_g C_g + 2D_{2g} C_{2g} - D_i C_i - D_v C_v^e \left\{ \exp \left[\frac{\Omega}{k_B T} \left(\frac{2\gamma_b}{r_b} - P \right) \right] \right\} \right) \quad (15)$$

On the other hand, the average radii of the loops and bubbles can be obtained by using the mean-size approximation [30,32]:

$$\langle (r_{il})^2 \rangle \approx \langle r_{il} \rangle^2 = \frac{S_i}{\pi b N_{il}} \quad (16)$$

$$\langle (r_b)^2 \rangle \approx \langle r_b \rangle^2 = \frac{3S_b}{4\pi N_b} \quad (17)$$

Substituting Eqs. (14)–(15) into above equations and with the aid of the chain's rule, the evolution of average radii is expressed as:

$$\frac{dr_{il}}{dt} = \frac{\Omega}{2\pi r_{il} b N_{il}} (k_{il}^2 (2D_{2i} C_{2i} + 3D_{3i} C_{3i}) + 2\pi r_{il} N_{il} (D_i C_i Z_{il}^i - D_v C_v Z_{il}^v)) + \frac{r_{il}^2 \Gamma_p N_b}{4r_{il} N_{il}} - \frac{r_{il}}{2N_{il}} \frac{dN_{il}}{dt} \quad (18)$$

$$\frac{dr_b}{dt} = \frac{\Omega}{4\pi r_b^2 N_b} (4\pi r_b N_b (D_v C_v + D_g C_g + 2D_{2g} C_{2g} - D_v C_v^e \left\{ \exp \left[\frac{\Omega}{k_B T} \left(\frac{2\gamma_b}{r_b} - P \right) \right] \right\}) - D_i C_i) - 2k_b^2 D_{2i} C_{2i} - 3k_b^2 D_{3i} C_{3i} + \frac{b}{8} \Gamma_p - \frac{r_b}{3N_b} \frac{dN_b}{dt} \quad (19)$$

It is obvious that the size of immobile sinks evolves by absorbing 3-D diffusional point defects and 1-D diffusional di- and tri-interstitials, as well as by loop punching, characterizing by their sink strengths. It should be pointed out that k_{il}^2 and k_b^2 are partial sink strengths of interstitial loops and bubbles to 1D migrating defects (2i, 3i) [52,63]

$$k_{il}^2 = 2 \times \frac{\pi r_{cd}}{2} \times 2\pi r_{il} \frac{N_{il}}{4} \left(\frac{\pi r_{cd}}{2} \left(\frac{\rho_d}{4} + 2\pi r_{il} \frac{N_{il}}{4} \right) + \pi r_b^2 N_b \right) \quad (20)$$

$$k_b^2 = 2 \times \pi r_b^2 N_b \left(\frac{\pi r_{cd}}{2} \left(\frac{\rho_d}{4} + 2\pi r_{il} \frac{N_{il}}{4} \right) + \pi r_b^2 N_b \right) \quad (21)$$

Till now, a reduced CD model consisting of a set of coupled ordinary differential equations (ODEs) is developed, which can track the evolution of both mobile and immobile defects in irradiated tungsten with and without the helium implantation.

2.3. Parameters and experimental data

To solve above ordinary differential equations, a large number of parameters in the Cluster Dynamics (CD) model need to be determined. Beneficial from the relatively abundant experimental data around 1000 K, all simulations will be performed at this temperature unless stated elsewhere. Accordingly, corresponding parameters used in the present work are summarized in Table 1. It can be seen that most of parameters are obtained from atomistic simulations and experiments, and only four parameters associated with the saturation density of voids/bubbles are fitted from experimental data in the temperature range of 300–1200 K.

Actually, we have summarized available experimental data of irradiated tungsten (W) at various irradiation temperatures and doses in Table A1 in the Appendix. It is clear that the void size increases, and the density decreases with increasing the temperature, which justifies the rationality of these fitted parameters. Moreover, the experimental data with the helium implantation are summarized in Tables A 2 and A3 as well, wherein the size and density of bubbles present similar temperature dependency to these of voids. We will present the simulation results obtained at different temperatures in the next section and compare with this available experimental data to validate the selection of the parameters and to show the capacity of the developed model.

3. Result

A set of rate equations in section 2 are solved using a MATLAB ODE solver (ode15s) [67]. The results in terms of defect concentrations, loop and void/bubble densities and their mean diameters of single crystal irradiated tungsten are exhibited in the following.

3.1. Effect of irradiation temperature without helium implantation

The evolution of mobile defect concentrations at 1000 K without helium implantation is shown in Fig. 3(a). Initially, all kinds of defect concentrations increase until 10^{-9} dpa as they continuously generated from the collision cascade. Subsequently, the concentration of single-interstitials decreases due to its reaction with other species such as the vacancy-interstitial recombination, whereas the concentration of vacancies keeps increasing because of its low diffusivity. The concentrations of di- and tri-interstitials reduce as well, as they contribute to the nucleation of interstitial loops (see Fig. 3b). After 10^{-2} dpa, the concentrations of all defect types reach a steady state, which means that the defect production is compensated by their loss to sinks.

The corresponding density evolution of interstitial loops and voids are shown in Fig. 3(b). It is seen that after a quick nucleation stage, the density of both loops and voids achieves a stable value ($\sim 10^{22} \text{ m}^{-3}$), which presents a good agreement with experimental data shown in Table A1. For interstitial loops, their density keeps increasing at the steady state (after ~ 0.1 dpa). This is since the average loop size

Table 1
Parameters of the cluster dynamics model for irradiated tungsten.

Parameter	Symbol	Value	Unit	Reference
Materials parameters				
Lattice Constant	a_0	0.316	nm	[5]
Atomic volume	Ω	1.5825×10^{-29}	m^3	[5]
Burgers vector	b	0.274	nm	[18]
Dislocation density	ρ	10^{12}	m^{-2}	[18]
Recombination radius (I-V)	r_{iv}	0.465	nm	[18]
Capture radius of dislocation	r_{cd}	1.5	b	[32,52]
Attempt frequency	ν_0	6×10^{12}	s^{-1}	[64]
Poisson's ratio	ν	0.25	—	
Shear modulus	μ	161	GPa	
Irradiation parameters				
Displacement damage rate	G^{NRT}	1×10^{-6}	dpa/s	[25]
Helium ion flux	ϕ_{He}	1×10^{20}	ions $\text{m}^{-2} \text{s}^{-1}$	[25]
Helium implantation rate ¹	G_g	1×10^{-9}	dpa/s	
Neutron flux	$\phi(E_n)$	1×10^{17}	$\text{m}^{-2} \text{s}^{-1}$	[18]
Neutron scattering cross-section for helium	$\sigma_s(a, He)$	1×10^{-28}	m^2	[18]
Surviving fraction	f	0.85	—	[17]
Fraction of interstitials produced as 2i	ϵ_{2i}	0.15	—	[65]
Fraction of interstitials produced as 3i	ϵ_{3i}	0.1	—	[65]
Fraction of vacancies produced as vacancy clusters ²	ϵ_v	0.15	—	[65]
Reference parameter of saturation density	N_{sat0}	5×10^{22}	m^{-3}	Fitted
Temperature-dependent parameters	P_{T1}	5	1	Fitted
Temperature-dependent parameters	P_{T2}	-4×10^{-3}	/K	Fitted
Implantation-dependent parameters	P_{G_g}	5.19×10^4	1	Fitted
Energies				
Formation energies of SIA, vacancy and helium	E_i^f, E_v^f, E_g^f	9.46, 3.80, 4.00	eV	[18]
Migration energy of SIA, di-interstitials and tri-interstitials	$E_i^m, E_{2i}^m, E_{3i}^m$	0.013, 0.024, 0.033	eV	[18,66]
Vacancy migration energy	E_v^m	1.66	eV	[18]
Migration energy of helium and di-helium	E_g^m, E_{2g}^m	0.06, 0.2	eV	[18]
Binding energy of di-interstitials and tri-interstitials	E_{2i}^b, E_{3i}^b	2.12, 3.02	eV	[64]
Bind energy of di-helium	E_{2g}^b	1.02	eV	[64]
Binding energy of HeV, He ₂ V-V, He ₂ V-He	$E_{gv}^b, E_{2gv}^b(v), E_{2gv}^b(g)$	4.57, 6.65, 4.10	eV	[5,18]
Diffusion pre-exponential factors				
SIA, di-interstitials and tri-interstitials	$D_{0,i}, D_{0,2i}, D_{0,3i}$	$8.77 \times 10^{-8}, 7.97 \times 10^{-8}, 3.92 \times 10^{-8}$	m^2/s	[18,66]
Vacancy	$D_{0,v}$	177×10^{-8}	m^2/s	[18]
Helium and di-helium	$D_{0,He}, D_{0,2He}$	$2.95 \times 10^{-8}, 3.24 \times 10^{-8}$	m^2/s	[18]
Bias factors				

Table 1 (continued)

Parameter	Symbol	Value	Unit	Reference
SIA loop – I, SIA loop – V	Z_{ii}^i, Z_{ii}^v	1.2, 1.0	—	[18]
Dislocation line – I, Dislocation line – V, Dislocation line – He, Dislocation line – He ₂	$Z_d^i, Z_d^v, Z_d^g, Z_d^{2g}$	1.2, 1.0, 1.0, 1.0	—	[18]
Bubble – V, Bubble – He, Bubble – He ₂ , Bubble – I	$Z_b^v, Z_b^g, Z_b^{2g}, Z_b^i$	1.0, 1.0, 1.0, 1.0	—	

¹ The helium implantation rate G_g is converted to the standard dose rate, while helium ion flux is obtained from the relevant references [25].

² The fraction of various produced defects are estimated from the OKMC simulation results with different helium irradiation energies [65].

decreases at this stage (see Fig. 3c as well), which results in an accelerated nucleation rate as mobile SIA clusters cannot be fully absorbed by loops.

Average defect sizes are shown in Fig. 3(c). Similarly, both types of defects present a sharp growth initially, and follow a steady state regime. However, it should be emphasized that the voids exhibit a higher sensitivity to one-dimensional migration defects (see Eq. (21) for their sink strength), i.e., 2i and 3i, compared to single vacancies (v) and single interstitials (i). As voids grow, they do not stop absorbing more 2i and 3i until reaching a critical size, which corresponds to the balance of interstitial and vacancy fluxes towards the voids. The absorption of many 2i and 3i by voids in turn results in a shrinkage of interstitial loops, which further induces a reduction of their sink strengths and an increase of the nucleation rate due to the accumulation of single interstitials. Similar phenomena have been observed in experiments [12], as well as other cluster dynamics studies [59].

We then perform a temperature-dependence study in Fig. 4 and compare the defect number density and the size with experimental data across a wide range of temperatures. Because small clusters cannot be observed by TEM, the interstitial loop density obtained from experiments lags behind that obtained by CD simulations [30]. However, the nucleation rate of interstitial loops in the simulation and experiments is almost the same (see slopes in Fig. 4(a)), indicating that the kinetic of interstitial loop nucleation is well captured by the model and is controlled by a diffusional process. It is also interesting to note that interstitial loops fully saturate at low temperatures, whereas at higher temperatures, the loop density keep increasing at the steady stage as discussed previously. Furthermore, a natural outcome of our CD model is that, due to the rapid increase in diffusion coefficients at high temperatures, the density of interstitial loops increases with rising temperature. Conversely, at low temperatures, the diffusion coefficient is less affected by changing temperatures, leading to a decrease in loop density. This phenomenon is fully consistent with experimental observations that the loop density first decreases then increases with increasing temperatures at the same dose level. As for voids, it is pronounced that rising temperature always induces a decrease in their density. In regard to the mean size of loops and voids shown in Fig. 4(b) and (d), it can be found that at low temperatures (300 K), void size remains almost zero, which shows a good agreement with the experiments that no void is observed at this temperature [9,68,69]. This further allows the continuous growth of interstitial loops by absorbing a big amount of mobile SIA defects. At high temperatures however, the size of interstitial loops begins to decrease after reaching a critical value. Instead, the size of voids increases with rising temperature, and it remains almost constant in the steady state regime as shown in Fig. 3. Moreover, it is worth to note that the average loop size decreases with increasing temperature in the studied temperature range. This is found to be a result of the temperature dependency of defects' diffusivity. First, the diffusivity of interstitials increases with increasing temperature, which thus results in a larger nucleation rate and a higher loop density. From Table 1, it can be

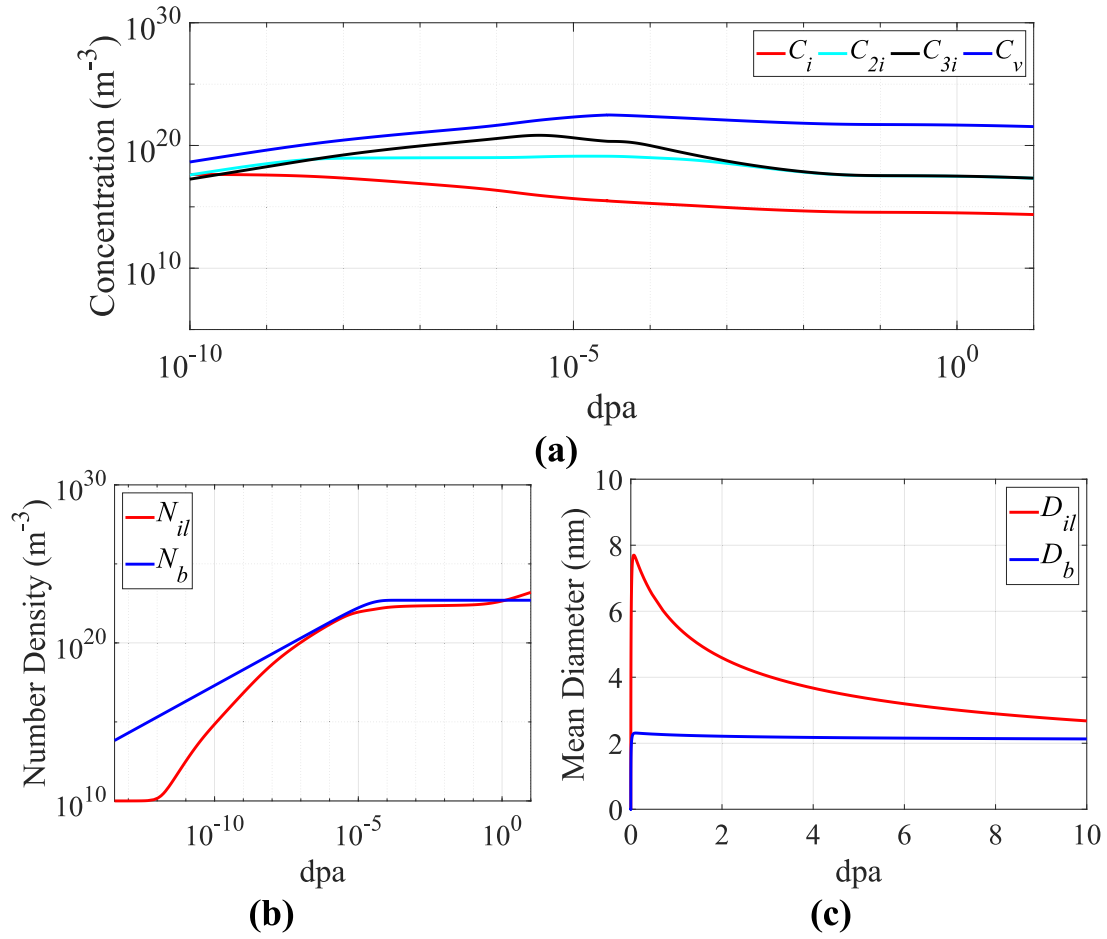


Fig. 3. (a) Concentrations of point defects, di-interstitials and tri-interstitials in single crystal tungsten at 1000 K, (b) Number density, and (c) mean diameter of interstitial loops and voids in single crystal tungsten at 1000 K.

also seen that temperature presents a stronger influence on the diffusivity of vacancies when compared to interstitials, which then promotes the absorption of vacancies by SIA loops and reduces the net interstitial flux towards loops with increasing temperatures. Both effects thus lead to a reduction in the average size of the dislocation loops with respect to temperatures. However, it should be pointed out that the effect of the loop coarsening is not considered in the simulations, which may cause an opposite size-temperature dependence at very high temperatures.

In conclusion, temperature primarily influences the diffusivity of mobile defects, which in turn strongly affect the immobile defect density and size. The distinct temperature-dependency for voids and loops qualitatively agrees with experimental observations [68,70–72]. And both number densities and sizes of voids and loops are quantitatively consistent with the experimental data at different doses and various temperatures from 300 K up to 1200 K. All these thus validate the choice of parameters and demonstrate the capability of the developed model.

3.2. Effect of helium implantation

In this section, we present the simulation results with the helium implantation at 1000 K and 1200 K using the same parameters listed in Table 1. It is worth noting that the loop punching is not considered in this section and thus the loop punching frequency Γ_p is set to be zero. With helium implantation, four additional mobile defects containing helium atoms need to be considered, which are He , He_2 , HeV , and He_2V . The concentration evolution of all mobile defects is shown in Fig. 5(a). Similar to the defect concentration evolution without helium implantation, all curves achieve a steady state after a fast transition regime.

However, it is noted that the stable value for vacancy concentration is slightly smaller than the former case due to the strong vacancy-trapping ability of helium. Then the number density and size of immobile defects are shown in Fig. 5(b) and 5(c) and compared to available experimental data. In general, both the density and size of helium bubbles are in good agreement with experimental data. Furthermore, compared with the non-helium case, the saturation concentration of bubbles is higher than the case in presence of the helium implantation. And more interestingly, although the size of interstitial loops performs a similar evolution to the non-helium case, the size of interstitial loops eventually drops to a value close to zero, indicating the annihilation of loops by absorbing vacancies. And the bubbles can continuously grow by absorbing helium atoms, leading to the net vacancy flux towards SIA loops. These observations are fully consistent with experiments with helium implantation as well, where the dislocation loops disappear above a certain irradiation doses (~ 4 dpa) but the size of bubbles slightly increases with doses [8,71–73].

In Fig. 5(c), it is important to note that some experimental data of dislocation loop sizes is larger than our simulation results. This is because: first, the higher energy of incident particles (~ 30 keV) may result in severer production bias, which induces a deviation of the cascade parameters from those used in our simulations; second, in experimental observations, only larger loops are statistically accounted for, which may overestimate the average loop size compared to our simulations. We have also compared the dislocation line density ($\sim N_{il}/D_{il}$) between simulations and experiments (the insert of Fig. 5(c)). The good consistency thus implies that our model can accurately predict the evolution of dislocation ensembles. Similarly for bubbles, the

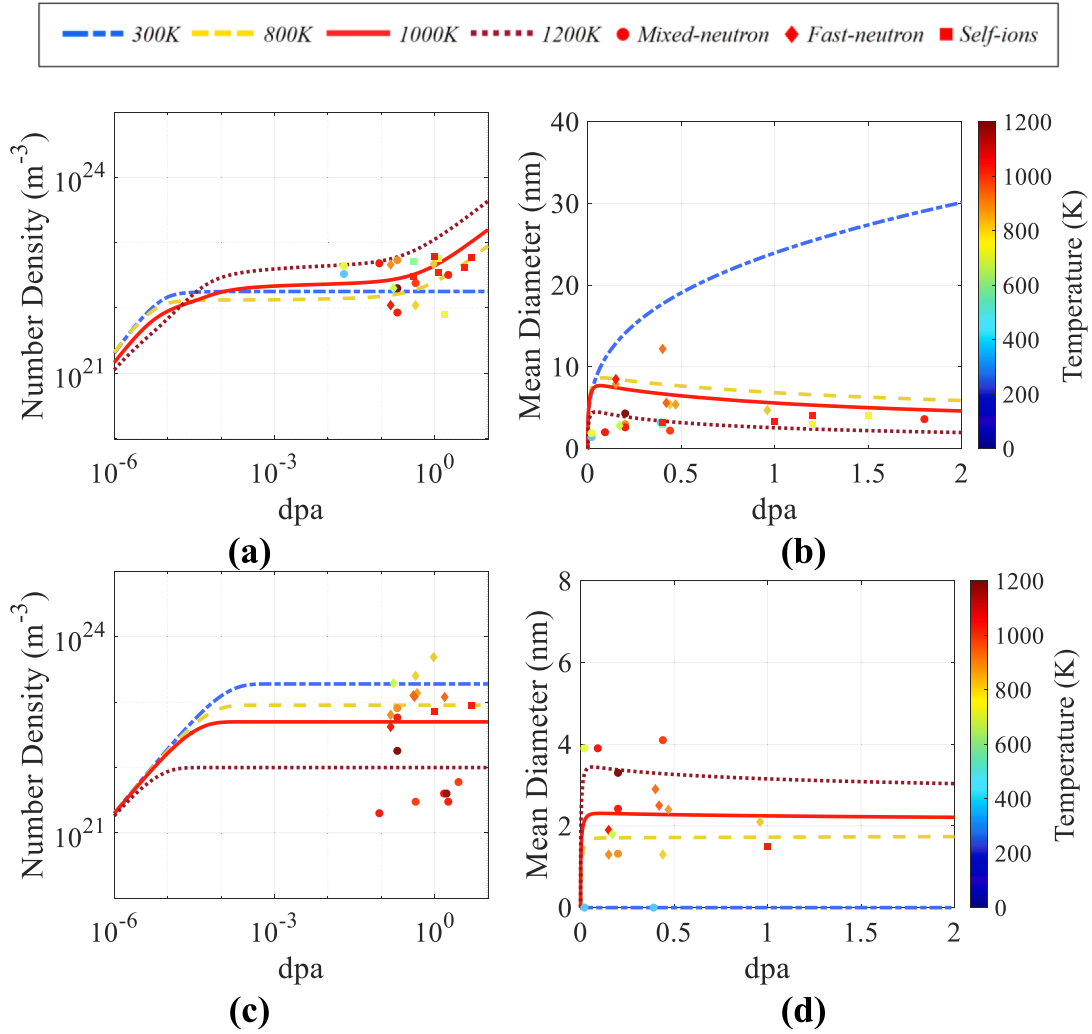


Fig. 4. Comparison between simulation results (lines) and experimental data (symbols) at different temperatures (300K, 800K, 1000K, 1200K) in terms of (a) SIA loop number density, (b) mean diameter of SIA loops, (c) number density of voids and (d) mean diameter of voids.

different radiation sources, energy and helium implantation rate may create the discrepancy between the model predictions and the experimental data.

3.3. Effect of loop punching mechanisms

At this stage, we have re-examined the above results. It is found that although the aforementioned model predictions in terms of the number density and the average size of loops and bubbles align with experimental results, bubbles in this case grow by preferentially absorbing helium atoms if their size achieves a critical value, which results in a very high helium-to-vacancy ratio and unrealistic internal pressure. This thus suggests that the loop punching needs to be incorporated to release the bubble pressure. As discussed in the introduction, many recent investigations have revealed that the loop punching mechanism plays a crucial role in the growth of helium bubbles. It has been highlighted that a helium bubble can relieve its internal pressure by emitting an interstitial loop, thereby increasing the vacancy concentration within the bubble and enlarging its size [15,73]. And depending on the bubble size, there may be two different ways for forming interstitial loops. For a small bubble, interstitial loops are formed by gathering and rearranging of the W self-interstitial atoms [15,74–76]. As for a large bubble, recent atomistic simulations [15] have unraveled that interstitial loops can be formed by directly emitting and cross-slipping of screw dislocations as illustrated in Fig. 6. It is found that when the pressure in an He bubble

risks to a critical level, a $1/2[111]$ screw dislocation can be generated on one of the 12 slip planes in tungsten. The subsequent cross-slips of such screw dislocation to adjacent slip planes ultimately results in the formation of a prismatic loop. As the shear stress is maximum at the bubble surface with an angle of 90 degrees, the emitted dislocation loops have a characteristic size of $r_{pil} = \frac{\sqrt{2}}{2}r_b$ [15]. And accordingly, once a loop punching is completed, the bubble will grow up by a size of $b/8$.¹ The purpose hereafter is therefore to account for this loop punching mechanism within our CD framework and examine its effect on the microstructure evolution.

Following the transition-state theory, the loop emission frequency appeared in Eqs. (11)–(15) can be written in an Arrhenius form as:

$$\Gamma_p = \nu_0 \exp\left(-\frac{E_a}{k_B T}\right) \quad (22)$$

where ν_0 is the attempt frequency (pre-factor), k_B being the Boltzmann's constant and T is the absolute temperature. E_a is a reference activation energy, which is expressed as [15]:

$$E_a = \frac{\mu b^2 r_{pil}}{2(1-\nu)} \left(\ln \frac{8r_{pil}}{\epsilon} - 1 \right) - n \left(4\pi r_b^2 \left(\frac{b}{8} \right) P \right) \quad (23)$$

¹ $\frac{4}{3}\pi(r_b + \Delta r)^3 = \frac{4}{3}\pi r_b^3 + \pi r_{pil}^2 b, r_{pil} = \frac{\sqrt{2}}{2}r_b \Rightarrow \Delta r = \frac{b}{8}$ shown in Fig. 6.

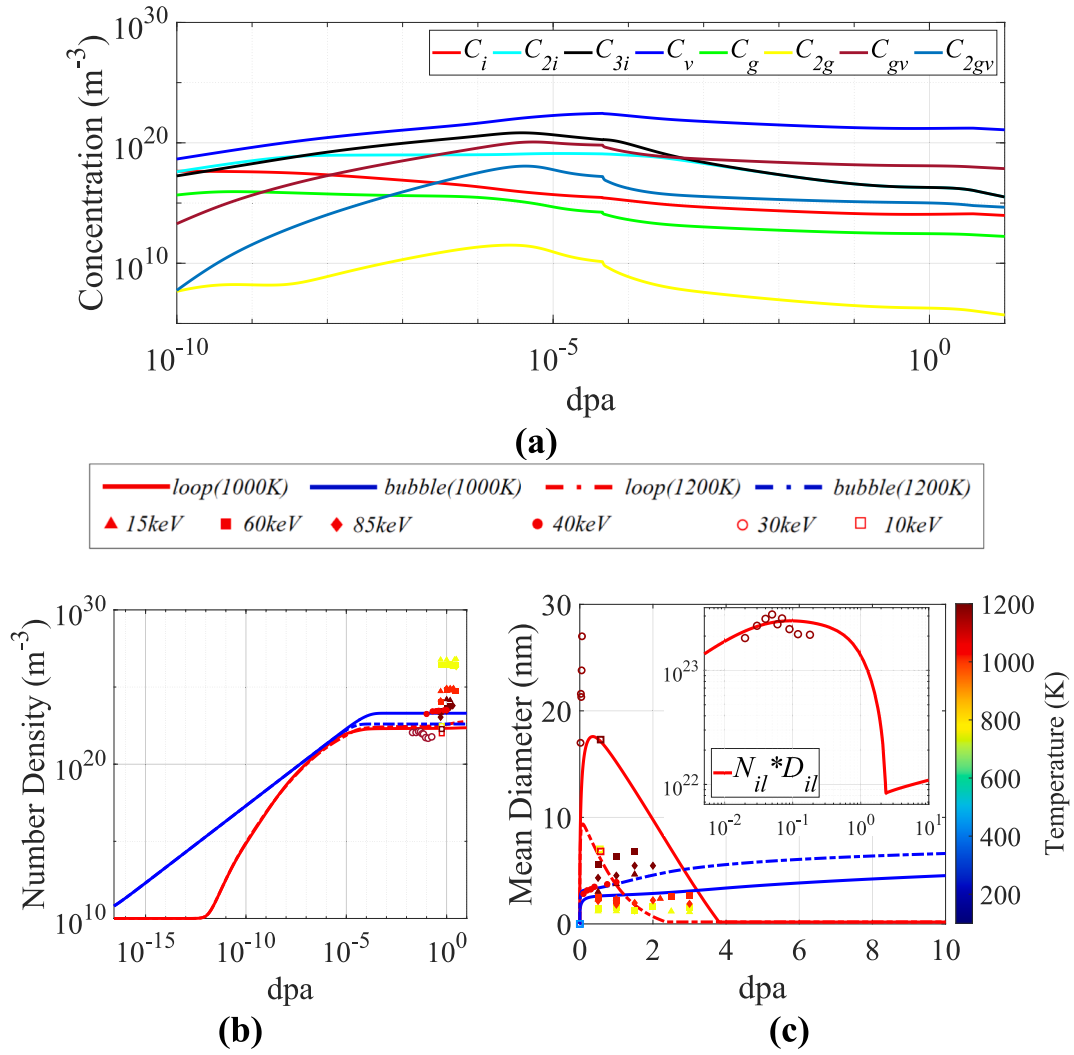


Fig. 5. Simulation results with helium implantation at 1000K and 1200K. (a) Concentrations of various types of defects in single crystal tungsten. (b) Number density and (c) mean diameter of interstitial loops (open symbols) and bubbles (filled symbols) from experiments and simulations (lines); the insert represents the dislocation line density at 1200K.

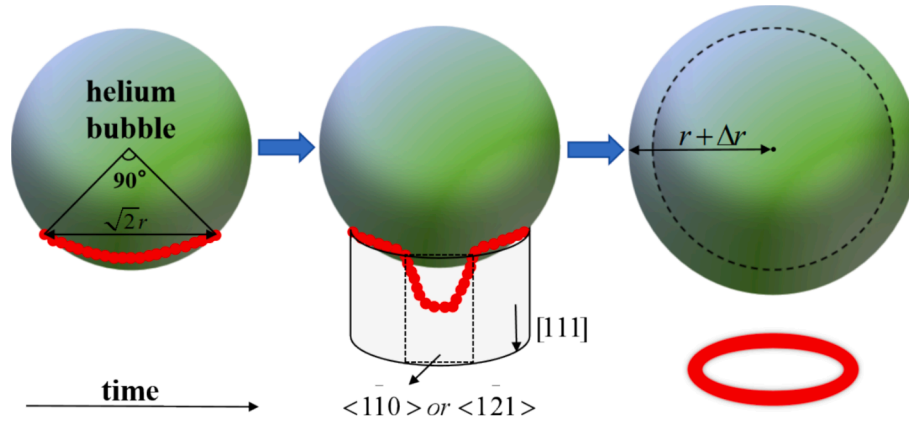


Fig. 6. A new loop punching mechanism observed in the atomistic simulations [15].

The above equation consists of two terms. The first term is the self-energy of the emitted dislocation loop, and the second term represents the work done by the internal pressure, which helps to push the loop out of the bubble. The parameter n is a pre-factor that regulates the

contribution of the internal pressure feedback term. Additionally, μ represents the shear modulus and ν denotes Poisson's ratio. b is the magnitude of the Burgers vector of the prismatic dislocation loop and r_{pil} is radius of the emitted interstitial loop. ϵ is the dislocation core size, which is usually chosen as $\epsilon \sim b$. P is the internal pressure, which is

expressed in equation of state (EOS) as [77]:

$$P = a_0(\rho) + a_1(\rho)T + a_2(\rho)T^2 \quad (24)$$

where $\rho = \frac{S_g}{S_v}$ is the helium-to-vacancy ratio, and $a_0(\rho), a_1(\rho), a_2(\rho)$ are empirical parameters from atomistic simulations [77]. It is noted that the EOS for iron and tungsten presents little difference as they have the same crystal structure and very similar material's properties [16]. The EOS for tungsten shown in [16] entails numerous parameters. We therefore adopt the EOS from iron for simplicity. In the following, we will show the effect of this loop punching mechanism and the influences of the activation energy on the loop growth kinetics.

The results considering the loop punching mechanism ($n = 0$) at 1000 K are shown in Fig. 7(a) and (b). In contrast to cases without loop punching, it is evident that the density of interstitial loops increases due to the loop emission by bubbles, which then leads to the decrease in the average loop size. It is also clear that the loop size starts to drop earlier. However, it should be noted that the initial drop is not only attributed to the net vacancy flux, but also the increase in the loop density. All SIA loops disappear eventually, but at a smaller dose compared to the loop punching free case (from 4dpa to 3dpa), which is better consistent with experiments. In regard to bubbles, they exhibit a faster growth rate only at the early stages, while the growth rate at the steady state is almost the same as the previous case without the loop punching. This transition can be easily explained in conjunction with the behavior of SIA loops. Initially, bubbles can grow up by emitting SIA loops, contributing to a rapid growth. However, the energy required for forming a loop increases dramatically with the growth of bubbles, which hinders the further loop emission. Moreover, once the SIA loops disappear, more di- and tri-interstitial flow towards bubbles, compensating the loop punching effect. Accordingly, the loop punching alone presents little influence on the size of bubbles. However, when we checked the average internal pressure and the helium to vacancy ratio of bubbles at this case (see Fig. 8), it can be seen that they are far beyond the realistic values, indicating the incompleteness of the model.

3.4. Effect of internal pressure feedback

To ameliorate the model, we incorporate the internal pressure feedback (second term in Eq. (23)) into the framework. Actually, the activation energy of loop punching depends on both the bubble size and the internal pressure. Although the bubble size increases, the accumulation of the high pressure inside bubbles significantly lowers the activation energy, making it possible to emit loops to alleviate the pressure [15,73]. Consequently, this term plays a key role in controlling the helium-to-vacancy ratio and the pressure of bubbles. Now, the corresponding results considering the pressure feedback with $n = 1$ are shown in Fig. 7(a) and (b). It can be observed that the consideration of the pressure feedback has no significant effect on the bubble size and bubble density. But in Fig. 8(a), it can be seen that the internal pressure feedback drastically lowers the required activation energy as expected, thereby greatly increasing the frequency of interstitial loop emission. This thus further results in a rapid increase in the density of interstitial loops and a sharp decrease in their size. Additionally, the helium-to-vacancy ratio in this case drops to a rational value as shown in Fig. 8(b) [10], confirming the effectiveness of the model.

A further validation is presented in Fig. 9(a), where we compare the internal pressure of bubbles as a function of their sizes from simulations, to the results obtained from MD and EOS estimations [16]. Combined with the bubble size evolution shown in Fig. 9(b), the kinetic behavior of bubbles can be categorized into three stages. The first stage is

characterized by a rapid growth, where helium bubbles primarily grow by absorbing vacancies, resulting in an almost zero pressure. In the second stage, helium bubbles predominantly grow by absorbing helium, while excess vacancies are absorbed by interstitial loops, causing a sharp increase in their internal pressure. In the third stage, once the size of helium bubbles and the internal pressure reach a certain threshold, the loop punching mechanism becomes pronounced, maintaining a constant pressure in bubbles as observed in MD. At the beginning of the stage three, the emitted loops are small due to the small size of bubbles, resulting in a sharp decrease in the average size of the loops. Subsequently, as bubbles grow and loops shrink, the shrinking rate of the interstitial loops diminishes. And eventually, the interstitial loops begin to grow slowly if the loops emitted by the loop punching is larger than the average size of loops as illustrated in Fig. 9(b).

From above discussions, it is evident that the loop punching mechanism plays a crucial role in the nucleation and evolution of interstitial loops and bubbles. The activation energy E_a is a key factor influencing the frequency of loop punching. Therefore, we can adjust E_a to mitigate the growth of bubbles and thus the irradiation-induced swelling. This can be achieved by changing the pre-factor n in Eq. (23). The results of bubble's density and size and helium-to-vacancy ratio with different n are shown in Fig. 10(a) and (b), respectively. It is seen that reducing the internal pressure feedback by using smaller n can effectively reduce the bubble growth rate at the steady state. And more importantly, the helium-to-vacancy ratio at this case is approximately 2, which is fully consistent with experimental measurement under the same irradiation conditions [10]. The present work thus illustrates the effect of the internal pressure of bubbles and highlight the significance of the loop punching mechanism in studying the microstructure evolution of irradiated tungsten.

4. Conclusion

We develop a reduced cluster dynamics model to study the microstructure evolution of irradiated tungsten, accounting for the migration and interaction of mobile defects, as well as the nucleation and growth of interstitial loops and voids/bubbles. The model has been adopted to various irradiation conditions, and the results in terms of the defect number density, average defect size and the helium-to-vacancy ratio are all in good agreements with experimental observations and molecular dynamics simulation results. The following key conclusions are drawn from this study:

The model demonstrates that voids and bubbles cannot form at low temperatures (below 300 K), while at high temperatures (above 1000 K), interstitial loops tend to disappear if the dose increases above 3dpa. These results are consistent well with experimental observations.

It is found that the density of interstitial loops increases and their size decreases with increasing temperatures, while the opposite trend is observed for voids/bubbles. And the helium implantation will enhance the nucleation and growth of vacancy-type clusters as helium prefer to bind together with vacancies.

The consideration of the loop punching can slightly accelerate the growth rate of bubbles and has a significant effect on the bubble's pressure and the helium-to-vacancy ratio.

The internal pressure feedback of the loop punching needs to be considered to match the pressure and the helium-to-vacancy ratio between simulations and experimental measurements. And this term can effectively adjust the growth kinetics of bubbles.

The current study is an important step towards the full development of rate-theory based models for predicting the irradiation damage in tungsten.

$$\begin{aligned} 2 \frac{dS_g}{dt} = & 4\Omega\pi r_b N_b (D_g C_g + 2D_{2g} C_{2g}) \frac{dS_b}{dt} = \pi \left(\frac{\sqrt{2}}{2} r_b \right)^2 b \Gamma_p N_b - 2\Omega k_b^2 D_{2i} C_{2i} - 3\Omega k_b^2 D_{3i} C_{3i} \\ & + 4\Omega\pi r_b N_b \left(D_v C_v - D_i C_i - D_v C_v \left\{ \exp \left[\frac{\Omega}{k_b T} \left(\frac{2r_b}{r_0} - P \right) \right] \right\} \right). \end{aligned}$$

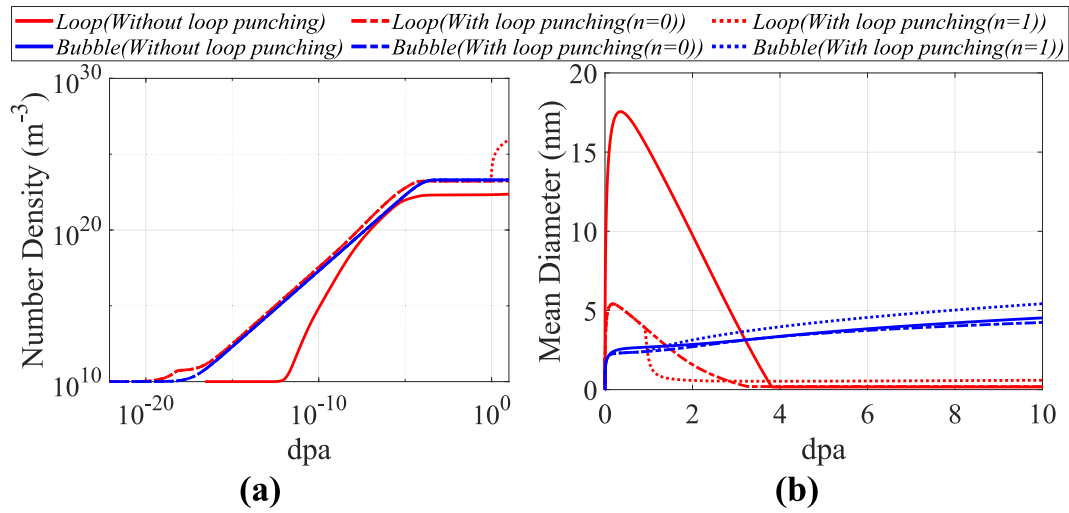


Fig. 7. Comparison of (a) number density, (b) mean diameter of interstitial loops and bubbles for different cases at 1000 K.

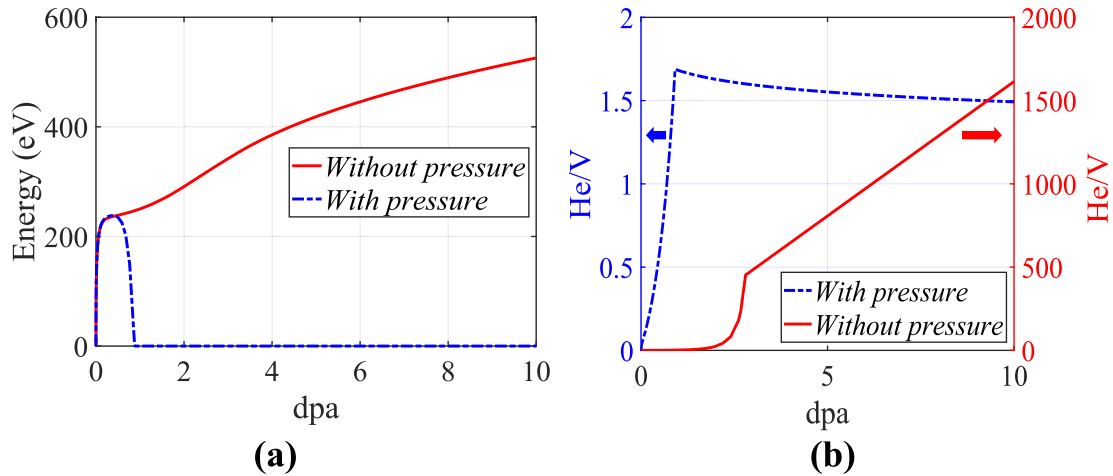


Fig. 8. Effect of the internal pressure feedback on (a) the activation energy of loop punching, and (b) the helium-to-vacancy ratio of helium bubbles.

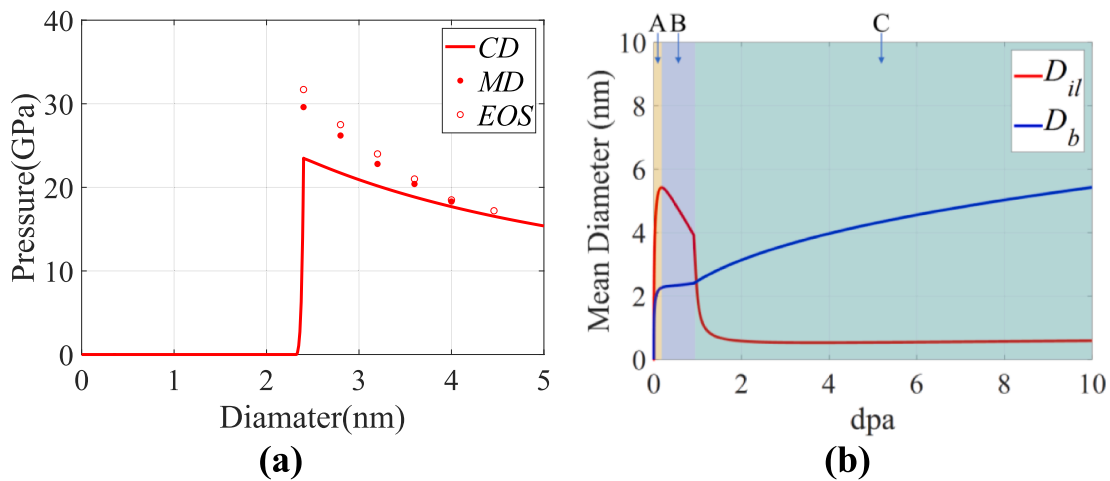


Fig. 9. (a) The evolution of the internal pressure of bubbles obtained from simulations, MD results [16] and EOS estimations [16]. (b) Bubble size evolution as a function of irradiation doses. Shaded areas highlight three different stages.

CRediT authorship contribution statement

Zhiyuan Li: Writing – review & editing, Writing – original draft,

Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. Gaowen Ding: Writing – review & editing, Methodology, Investigation, Conceptualization.

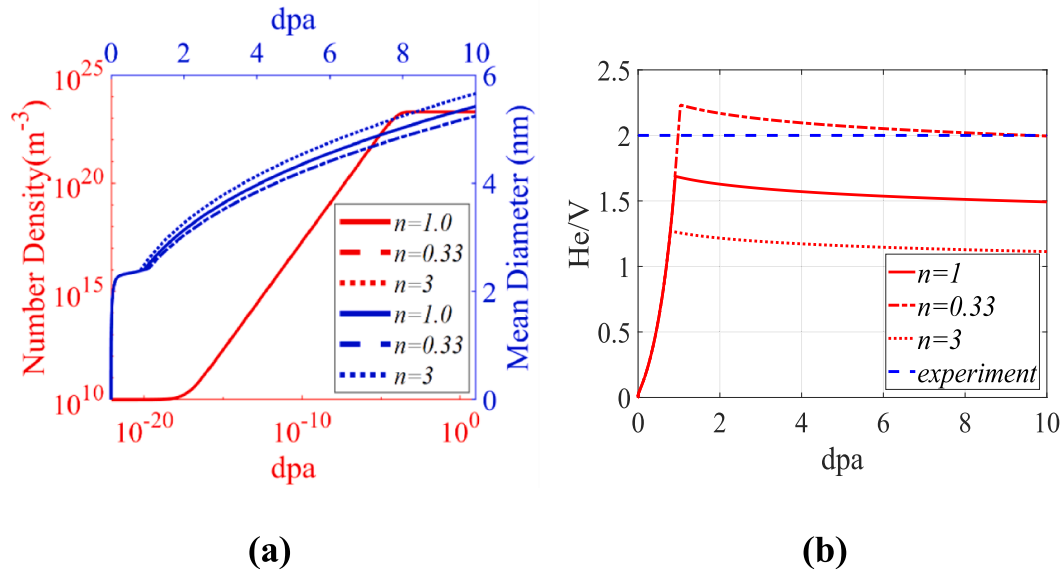


Fig. 10. The effect of pre-factor n on the evolution of (a) nucleation (red line) and growth (blue line) of bubbles, and (b) helium-to-vacancy ratio.

Pengchuang Liu: Writing – review & editing, Supervision, Project administration, Funding acquisition. **Tenglong Cong:** Writing – review & editing, Supervision, Project administration. **Yang Li:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Table A1 summarizes the density and size of voids and loops observed in irradiated tungsten under different temperatures and doses for neutron irradiation and self-ion irradiation. And the data using helium ion sources are presented in Table A2. Table A3 summarizes the experimental data of the density and size of bubbles and loops in tungsten with both helium ion irradiation and mixed ion irradiation. Notably, the helium bubble density provided by Harrison et al. [78] is in area density. We converted these data into number densities, which may result in an overestimation since their helium bubble size is relatively small. These experimental data are collected from Transmission Electron Microscopy (TEM) observations, which is however not able to distinguish very small defect clusters, i.e., clusters smaller than 1.0–1.5 nm [79,80]. Even using Positron Annihilation Spectroscopy (PAS) cannot accurately measure the small irradiation defects. Therefore, all data presented in Tables A1–A3 have a mean diameter larger than 1 nm.

Table A1

Summary of density and size of voids and loops at different irradiation temperatures and doses for neutron and self-ion irradiation.

Irradiation source	T [K]	Dose [dpa]	Void density [$\times 10^{22}/m^3$]	Void diameter [nm]	Loop density [$\times 10^{22}/m^3$]	Loop diameter [nm]
Mixed-neutron (HFIR) [7]	363	0.02	–	–	3.3	1.4
	363	0.39	–	–	7.2	3.2
	733	0.02	0.3	3.9	4.4	1.9
	1070	0.092	0.2	3.9	4.8	2.0
	973	0.44	0.3	4.1	2.4	2.2
	1043	1.80	0.3	6.3	3.2	3.6
	1073	1.50	0.4	5.6	–	–
	973	2.80	0.6	5.8	–	–
Mixed-neutron (HFR) [7]	1173	1.67	0.4	5.0	0.01	20
Mixed-neutron (BR2) [7]	873	0.2	8.1	1.32	5.38	2.98
	1073	0.2	5.81	2.42	0.85	2.60
	1473	0.2	1.80	3.30	2.0	4.30
Fast-neutron (JOYO) [7]	873	0.15	6.4	1.3	4.6	7.9
	1073	0.15	4.2	1.9	1.1	8.5
	673	0.17	19.5	1.8	2.0	2.8

(continued on next page)

Table A1 (continued)

Irradiation source	T [K]	Dose [dpa]	Void density [$\times 10^{22}/\text{m}^3$]	Void diameter [nm]	Loop density [$\times 10^{22}/\text{m}^3$]	Loop diameter [nm]
2 MeV self-ions [68]	804	0.44	25.3	1.3	1.1	5.4
	811	0.96	49.0	2.1	4.7	4.7
	856	0.47	13.8	2.4	0.2	5.4
	913	0.40	12.7	2.9	0.3	12.2
	923	1.54	12.0	4.7	—	—
	929	0.42	12.1	2.5	0.1	5.6
	573	0.4	—	—	5.1	2.9
	773	1.2	—	—	5.8	3.0
	773	1.5	—	—	0.8	4.0
	1023	0.4	—	—	3.0	3.2
	1023	1.2	—	—	3.5	4.1
	1023	3.6	—	—	4.1	3.7
18 MeV self-ions [81]	1073	1.0	7.1	1.5	6.1	3.3
	1073	5.0	8.9	1.6	6.0	3.6

Table A2

Summary of density and size of bubbles under different helium ion irradiation.

Irradiation source	He ions [keV]	T [K]	Dose [dpa]	He fluence [$10^{21}/\text{m}^2$]	bubble density [$10^{23}/\text{m}^3$]	bubble diameter [nm]
Helium ions [78]	15	773	0.5	0.18	8409.1	1.19
			1.0	0.36	8479.8	1.20
			2.0	0.72	7700.6	1.08
			3.0	1.1	7985.8	1.12
		1023	0.5	0.18	283.8	2.31
			1.0	0.36	327.5	2.36
			2.2	0.79	467.7	2.37
			2.2	0.79	467.7	2.37
		1273	0.5	0.18	19.2	3.02
			1.0	0.36	225.4	3.75
			1.5	0.54	301.9	4.65
			1.5	0.54	301.9	4.65
	60	773	0.5	0.4	5813.4	1.48
			1.0	0.8	6009.8	1.53
			1.5	1.2	4988.5	1.27
			2.0	1.6	6402.6	1.63
		1023	3.0	2.4	6088.3	1.55
			0.5	0.4	65.5	2.45
			1.0	0.8	308.1	2.18
			2.5	2.0	386.1	2.57
			3.0	2.4	395.6	2.69
		1273	0.5	0.4	87.6	5.57
			1.0	0.8	185.1	6.28
			1.5	1.2	283.6	6.79
			1.5	1.2	283.6	6.79
		773	1.5	1.6	4144.4	1.24
			3.0	3.2	5243.7	1.56
		1023	0.5	0.53	56.8	2.15
			1.0	1.07	253.2	1.80
			1.5	1.6	293.2	1.96
			2.0	2.13	319.4	2.21
		1273	3.0	3.2	—	1.88
			0.5	0.53	20.9	4.32
			1.0	1.07	61.1	4.53
			1.5	1.6	152.9	5.46
			2.0	2.13	186.0	5.46
Helium ions [11]	40	1073	0.1	0.07	1.8	2.82
			0.2	0.14	2.7	3.14
			0.3	0.22	2.8	3.24
			0.4	0.29	2.9	3.48
			0.75	0.55	3.1	3.72
			1.0	0.73	3.2	3.92
			1.0	0.73	3.2	3.92
			1.0	0.73	3.2	3.92

Table A3

Summary of density and size of bubbles and loops under helium ion and mixed-ion irradiation conditions.

Irradiation source	T [K]	Dose [dpa]	bubble density [$10^{23}/\text{m}^3$]	bubble diameter [nm]	Loop density [$10^{22}/\text{m}^3$]	Loop diameter [nm]
10 keV He ⁺ [82]	773	0.17	4.8	1.81	—	—
		0.34	6.1	2.0	—	—
		0.45	7.3	2.0	—	—

(continued on next page)

Table A3 (continued)

Irradiation source	T [K]	Dose [dpa]	bubble density [10 ²³ /m ³]	bubble diameter [nm]	Loop density [10 ²² /m ³]	Loop diameter [nm]
30 keV H ₂ ⁺ and He ⁺ dual-beam[12]	1073	0.57	6.5	2.1	3	7.0
		0.45	2.6	3.5	—	—
	1273	0.57	3.8	3.3	1	6.8
		0.17	7.3	3.3	—	—
		0.34	17.1	3.5	—	—
		0.45	23.4	4.1	—	—
	1173	0.57	28.0	4.0	1.9	17.3
		0.02	—	—	1.13	17.0
		0.03	—	—	1.14	21.6
		0.04	0.92	2.00	1.33	21.3
		0.05	—	—	1.30	23.8
		0.06	1.0	2.08	0.94	27.0
		0.07	1.1	2.24	0.89	32.1
		0.09	1.2	2.58	0.53	43.4
		0.12	1.3	2.78	0.48	43.3
		0.18	1.3	2.93	0.59	34.8
		0.26	1.3	3.02	—	—

Data availability

Data will be made available on request.

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